Holonet Demonstrator: the Parity Control Arm

Turning the $W(3,3)$ Contextuality Test into a Two-Arm Discriminator

Wil Dahn | `github.com/wilcompute/W33-Theory`

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Abstract

The Holonet Demonstrator Protocol v1 measures the contextual fraction of the $q = 3$ Witting fabric and predicts CF $= 1/10$. This note adds the missing *positive control*. The parity law — a generalized quadrangle $W(q)$ has an ovoid, hence a noncontextual model, if and only if q is even — predicts that the *same class* of single-photon apparatus, re-pointed at an *even*-order analyzer map ($W(2)$ with 15 rays or $W(4)$ with 85 rays), must read CF $= 0$. Running both arms turns a one-sided test ("observe 1/10, or kill") into a *discriminating* measurement: contextuality must appear on the odd fabric and vanish on the even fabric, with the contrast fixed by the geometry, not by any tunable parameter. A systematic that could fake a violation would have to respect the parity of the field order to survive both arms. The control arm ships with an explicit noncontextual model: the ovoid assignment constructed and verified in `analysis/w33_ovoid_construct.py`.

1 The two arms

The substrate is the symplectic generalized quadrangle $W(q) = \mathrm{GQ}(q,q)$, whose points are the measurement rays and whose totally-isotropic lines are the $n = (q+1)(q^2+1)$ measurement contexts. The parity law (witness `analysis/w33_audit_qscan.py`) gives the contextual fraction in closed form,

$$\mathrm{CF}(q) = \begin{cases} 0, & q \text{ even}, \\ \dfrac{1}{q^2+1}, & q \text{ odd}, \end{cases}$$

so the experiment has two arms with opposite, geometry-forced predictions:

Arm	Fabric	rays n	contexts	predicted CF
positive (v1)	$W(3)$	40	40	1/10
control	$W(2)$	15	15	0
control	$W(4)$	85	85	0

On the even-order arm an *ovoid* exists: a set of $q^2 + 1$ rays (five for $W(2)$, seventeen for $W(4)$) meeting every context exactly once. Assigning the value 1 to those rays and 0 to all others is a single, context-independent truth assignment that satisfies all n contexts simultaneously — a working noncontextual hidden-variable model. Its existence is precisely why the even arm must measure no contextuality.

2 The explicit control model

The control prediction is not an assertion; it is a constructed object. The witness `analysis/w33_ovoid_construct.py` builds the ovoid for $q = 2$ and $q = 4$ and verifies it three independent ways: it has size $q^2 + 1$, it meets every context in exactly one ray, and its rays are pairwise non-collinear (a cap). The same witness confirms that the odd order $q = 3$ admits *no* such assignment (the best leaves four of forty contexts unsatisfiable), which is the obstruction the positive arm detects. The explicit $W(2)$ ovoid is

$$\{0100,\ 1000,\ 1101,\ 1110,\ 1111\},$$

five rays, one per context; the $W(4)$ ovoid has seventeen rays and is written to `data/w33_ovoid_construct.json`.

3 Two notions of contextuality (what the control arm does and does not test)

Two exact results sharpen the scope of the control prediction, both computed in the repository.

First, *the even fabric is not globally classical.* Labelling the fifteen points of $W(2)$ by the fifteen nontrivial two-qubit Pauli operators, every line is a commuting triple whose product is $\pm I$, and the noncontextual ± 1 assignment system is *unsatisfiable*: the witness `analysis/w33_doily_mermin.py` computes the fifteen signs as 4×4 matrix products, proves unsatisfiability by exact F_2 elimination, and extracts the minimal certificate — six lines, every point covered an even number of times, an odd number of $-I$ lines: a Mermin–Peres magic square inside the doily. So $W(2)$ is *sign*-contextual. What the parity law kills on even q is specifically the *selection* contextuality — the exactly-one-click-per-context excess that the contextual-fraction estimators measure — for which the ovoid is a perfect noncontextual model. The $q = 3$ fabric is contextual in *both* senses. The two-arm contrast is therefore a statement about one declared observable, the CF statistic, and both arms must measure exactly that statistic.

Second, *the control arm provably needs a different measurement class.* A complete-basis realization of $W(q)$ maps each context to an orthonormal basis, hence lives in $\mathbb{C}^{q+1}$. For $q = 2$ this is impossible: a non-collinear pair of rays spans two dimensions of $\mathbb{C}^3$, leaving a one-dimensional orthocomplement that contains exactly one ray, while the geometry demands $\mu = 3$ distinct common neighbours there (`analysis/w33_realization_dimension.py` verifies the counts). So the doily cannot be interrogated by the $q = 3$ arm's complete von Neumann measurements in its matching dimension; its exactly-one-click statistic must be realized by a different measurement class, as the honest-scope box below already concedes. The same counting at $q = 3$ poses no obstruction — and the Witting realization exists — so $q = 3$ is the *smallest* order the one-photon $\mathbb{C}^4$ apparatus can realize at all, and by the parity law the smallest contextual one: two independent forcings of the same order.

4 Statistics and decision rule

Each arm uses the same raw-shot schema, validator, and estimator as v1. Distinguishing a contextual fraction p from 0 at $k\sigma$ needs, in the binomial planning approximation,

$$N \geq \frac{k^2\, p(1-p)}{p^2},$$

so the positive arm targets $N \approx 225$ valid coincidences for 5σ at $p = 1/10$. The control arm instead *bounds* CF *from above*: it must show no statistically significant violation of the noncontextual budget on the even fabric, i.e. an estimate consistent with 0 to the same few- percent resolution. Because the control arm has no contextual signal to integrate, its budget is set by the upper-confidence target rather than by a violation; a comparable detected-event count suffices to exclude CF $\gtrsim$ a few percent.

Decision rule

- **Pass (theory survives):** the $W(3)$ arm gives CF consistent with 1/10 *and* the $W(2)/W(4)$ arm gives CF consistent with 0.
- **Kill:** contextuality appears on an even-order fabric, or the odd fabric fails to show 1/10 — either breaks the parity law that the substrate identification rests on.
- **Inconclusive:** the two arms are run on apparatus whose analyzer imbalance or loss profile differs enough that the contrast cannot be attributed to the fabric; re-run with a shared calibration.

Honest scope

The control arm requires a different analyzer bank: the $W(2)$ and $W(4)$ ray families (15 and 85 rays) are not the 40 Witting tetrads, so this is a generalization of the v1 apparatus class, not a re-run of the identical optics. The claim is that the same *kind* of single-photon path/polarization-plus-qutrit setup realizes all three fabrics. Building the even-order analyzers is the engineering cost of buying the control. As in v1, a positive odd-arm result certifies only the finite contextual resource of the substrate, not the full physics program; the value added here is that the even-arm null makes a faked contextuality far harder, because the apparatus must reproduce a parity-dependent contrast it was never told about.

5 Repository command path

```
python analysis/w33_audit_qscan.py          # parity law: CF=0 (even) vs 1/(q^2+1) (odd)
python analysis/w33_ovoid_construct.py      # explicit even-q ovoid = the control model
python analysis/holonet_control_arm.py      # both arms through the unmodified estimators
python analysis/w33_doily_mermin.py         # sign vs selection contextuality, separated
python analysis/w33_realization_dimension.py # q=2 has no complete-basis realization
holonet audit                               # pin the architecture ledger to the same commit
```

The first line establishes the prediction, the second constructs the noncontextual model the control arm must reproduce, and the third synchronizes the executable specification with the physical run.

Companion to: `holonet_demonstrator_protocol_v1.tex`. **Primary witnesses:** `analysis/w33_audit_qscan.py`, `analysis/w33_ovoid_construct.py`.